

Cumulative Reflection

Over my three years at Iowa State in the College of Engineering, I have had the pleasure of taking many classes that pushed my engineering abilities. An important part of any experience is pausing to reflect at crucial moments, such as when completing a degree program before moving on to the rest of my life and career.

The Cyber Security Engineering program at Iowa State is evidently designed to create well-rounded engineers. I have been educated to evaluate problems holistically, appreciating multiple viewpoints to increase diversity and improve the solutions I develop. Additional exposure to group projects and a team-based capstone project has prepared other students and me to collaborate with colleagues, follow ethical practices, and understand large-scale project work.

A key part of my education was learning beyond the metaphorical walls of my courses. Iowa State offers many resources for devoted students who want to go further. During my first year at the university, I spent lots of time exploring, including the library. I found an exciting section containing many books on cybersecurity and information security. Over the last few years, I have returned to this section many times to pick up a book and read for a few minutes. Learning something new during this time was a great resource in class, as my prior understanding of introductory content gave me a small knowledge advantage over my classmates. It also helped me make intellectual connections across disciplines within computing, improving my knowledge retention in courses.

A strong education outside the classroom is a critical piece of a well-rounded education. One of my most formative experiences beyond the classroom at Iowa State was participating in the College of Engineering's Boeing Research Fellowship, which allowed me to do paid research under a professor in my field of interest: cybersecurity. I worked with Ph.D. candidate Mahesh Bhat and Dr. Shana Moothedath, focusing on security measures for energy-efficient 6G wireless networks. Through this experience, I gained knowledge I likely would not have obtained in a class, as it was a very specific topic. This knowledge benefited me in other courses and work projects during my internship, as we worked with networks or artificial intelligence. This is one of the most convincing reasons to continue lifelong education. Continuously learning new things gives you a broader perspective on problems and a wider range of knowledge, which is incredibly valuable in industry. I have also expanded my knowledge by earning CompTIA cybersecurity certifications, adding them alongside my degree to demonstrate more specific expertise.

One of the biggest challenges Iowa State prepared me for was starting my internship, which turned into a part-time job. Many people acknowledge the gap between classroom and work experience, but I feel my education prepared me well for my career. I was inexperienced with many of the industry tools, but I was taught to handle problems holistically. With this perspective, I consider what I know and how I can apply it to my current situation so I can thrive. Additionally, I was given a project to redevelop my company's data loss prevention program because what they had was lackluster. I knew this was a large project to take on; however, I followed the engineering design process and completed it successfully. I analyzed the situation, researched solutions, gathered additional information, and considered corporate needs, then created an executive summary of the implementation plan to propose to senior leadership. The plan was approved, and we moved to implementation. When possible, I will monitor the implementation's success and identify areas for improvement. My courses at Iowa State, which required me to use the engineering design process, prepared me to attack this project.

If I could redo my college experience or advise my younger self, I would emphasize the importance of exploring many fields of cybersecurity early on. Because I lacked a broad knowledge base early on, I struggled to choose a topic for my Master's thesis. Additionally, it would provide a stronger foundation to make success in my courses more easily achievable. Additionally, I would work to make more friends within my degree program. Having more friends in my courses would make them more enjoyable and provide other resources for studying. Although I am pointing out things I would change, I am very happy with the majority of my experience.

My high school learning style was definitely not suited to college. I was naturally gifted as a child: topics came easily to me, and I seldom had to study. This changed completely in college as topics were more challenging, faster-moving, and more specialized. One example of this was my research on Low-Rank Adaptation for parameter-efficient transfer learning for encrypted network traffic in 6G networks. I had essentially no knowledge of this topic when I began. To gather as much information as possible from credible sources, I spent many hours reading published, highly cited research papers on transfer learning methods, machine learning, network traffic identification, and related topics. With that information, I worked with my mentors to decide which path to take for our project and began prototyping what we wanted. I worked through many models before finding one that worked best; however, this gave me more experience with the broader subject. Throughout the iterations, I researched as needed to learn specific topics or model structures. This defines my learning style of experiencing – I best absorb knowledge when I can apply it and base new knowledge on a necessity.

A crucial part of learning is applying your knowledge, whether on an exam or in a practical, to prove mastery of the content. In my Cybersecurity Tools and Concepts course, I learned primarily about penetration testing – also referred to as ethical hacking. This course gave me the theoretical knowledge to apply in this field. This past summer, I took the CompTIA PenTest+ certification exam, which allowed me to apply this knowledge in practical scenarios. I passed on the first try by applying the concepts I learned to logical problems. Nearly every problem could be solved strategically by applying separate pieces of knowledge to narrow options down to a specific answer or exploit a specific vulnerability.

Over the last three years, as mentioned, my learning strategies have developed as my courses required. In high school, my academic skills were much less developed; I didn't take notes, I finished homework in class while the teacher taught, and I didn't need to study for exams. In my first year, I had to build strong time-management skills as I took 18 credits and pursued major leadership opportunities outside of academia. I used university resources and mapped out my time in Shovel to ensure I had enough time to complete assignments, study for exams, and still make time for myself. Beyond that, I had to learn to take better notes to recall information in all of my classes. I began by handwriting notes, which progressed to typed notes alongside transcribed audio recordings. I found these worked well, but they also let me stop focusing in class, which reduced my learning. I have since returned to detailed typed notes, which force me to stay attentive in class while still giving me strong resources to rely on when needed. Doing this has evidently improved my grade and decreased the amount of studying, or rather reteaching, necessary for each exam I take.

As I complete my undergraduate degree, I must consider how I will continue my education to ensure my success in my chosen field. My first goal is to obtain my Master of Science degree in Cybersecurity. This will give me a strong foundation to succeed in industry or pivot to a doctoral degree to work in research or academia. Beyond this, I plan to continue earning industry certifications that require me to acquire additional knowledge in niche fields of cybersecurity. This will push me to keep broadening my knowledge and improve my attractiveness as an employee. While I work on my M.S., I intend to pursue a cybersecurity analyst certification. This will also support my professional development goals at work.

Over three years in Iowa State University's Cyber Security Engineering program, I evolved from a novice in the engineering field to a disciplined problem-solver. By taking advantage of opportunities like my internship and research fellowship, I further strengthened my academic work by giving myself a way to apply my knowledge. Beyond direct knowledge, my courses and projects prepared me to apply the engineering design

process to problems I encounter following graduation to develop logical solutions. With all of this, I believe I am fully prepared to enter the cybersecurity industry as a professional.